



UNIVERSITY OF
CENTRAL FLORIDA

EEL 3926L - JUNIOR DESIGN

Section: 0012

*College of Engineering and Computer
Science*

Department of Electrical and
Computer Engineering

Course Information

Term: Fall 2026

Class Meeting Days: W

Class Meeting Time: 09:00AM - 11:50AM

Class Meeting Location: ENG1 O274

Modality: P

Credit Hours: 1.00

Instructor Information

Name: Avra Kundu

Office Location: ENGI - 118

Office Hours

M: 12:00PM-2:00PM

Email: Avra.Kundu@ucf.edu

Name: Arthur Weeks

Office Location: HEC-205/ ENGI-456

Office Hours

M,T,TH: 2:00PM-4:00PM W: 3:00PM-5PM

Email: Arthur.Weeks@ucf.edu

Office Hours

Dr. Weeks: T,TH: 2:00PM-4:00PM W: 3:00PM-5PM

Dr. Kundu M: 12:00pm - 2:00pm.

Course Description

EEL 3926L ENGR-ECE 1(0,3)PR/CR: Complete the following:EEE3307C - Electronics I (4)EEL4742C - Embedded Systems (3)ECE Design Lab: Preparatory skills for Senior Design: Engineering design requirements, test equipment utilization, microcontroller usage, schematic capture, PCB design tools, PCB assembly using components and reflow soldering. Fall, Spr, Sum.

Introduction to project definition through development of appropriate engineering design requirements specifications. Analog hardware design, test equipment utilization, micro-controller development tool usage, schematic capture and PCB design tools. Students design, implement and test projects.

Required Course Materials and Resources

Laboratory Project Manuals

Authors: Electrical and Computer Engineering

Publication Date: Fall 2025

Notes: Available via webcourses for EEL3926

Student Learning Outcomes

A student undertaking this course will be able to carry out the following tasks: Design PCB, implement and test electrical engineering projects, fabricate a PCB and write embedded system software.

Course Assessment and Grading Procedure

1. Syllabus, solder training, BOM for new msp430 project, and Fusion license registration. - **5%**
2. TI WEBENCH, introduction to regulators and the testing of regulators. The student will prototype a 3.3 volt and a +5 volt regulator and an introduction to regulators. - **10%**
3. The student will design and layout on a PCB using Fusion these two regulators and order their blank PCB's. - **10%**
4. The students will research and design a schematic for the course project using MSP430g2553 datasheet and relevant peripherals' documents. - **5%**
5. The student will design PCB board layouts for their junior design msp430 project. The students will order the PCB for their project. - **15%**
6. The students will prototype their junior design msp430 project using the MSP430g2553 ET dev board. - **15%**
7. The students will continue with software for their junior design msp430 project. - **5%**
8. The students will build and test their regulators. **5%**
9. The students build their PCB and test their project. - **20%**
10. Attendance. - **10%**

Grading Scale

Letter Grade	Percentage
A	90 -100%
B	80 - 89%
C	70 - 79%
D	60 - 69%
F	0 - 59%

Communication Expectations

TBD

Policies for Course Grade

Assignments:

- Assignment will be given via Webcourses@UCF.
- All assignments due date be posted / announced when the assignment is given.
- Every student needs to submit an independent assignment solution **(no group projects)**.
- This course may require additional time beyond the fixed that which is scheduled. Additional laboratory time will be scheduled as needed.
- Note that you cannot expect to learn and pass this course without working and understanding the assignments and attending lectures.

Midterm Exams, Quizzes and Final Exams:

- **There are no Midterm Exams, Quizzes and Final Exams.**

Make-up of Assignments:

- No make-up Assignments are only allowed. for special circumstances. **No personal exemptions will be given. Only university pre approved activities or a documented medical emergency are allowed as an exemption to this rule.**

Any issues with grades that are received must be discussed with Dr. Weeks no later than one week after the grades are posted. After this time no grades will be changed. All email must be done through Webcourses.

Lectures:

All lectures will be conducted face to face in the mentioned classroom ENGR1.

Attendance/Participation:

All lectures / laboratory projects will be conducted face to face in the mentioned classroom. All students are expected to attend all classes in person in the classroom. Students are also expected to be on time and remain until class / laboratory is dismissed by the instructor.

Attendance is required for this class / laboratory as to complete the projects successfully. The students needs to be present during their assign lab section and performing the lab assignment. Attendance makes up 10% of the grade. Students are allowed to miss two laboratory attendances, a third missed attendance will result in one letter grade deduction.

Missing four classes / laboratory results in an F grade

Why an F grade if a student misses a total of 4 classes during the semester? This is equivalent of missing 1/4 of the semester.

Standard Wording for Lab Accessibility: "To add some flexibility to the attendance rule, based on availability students can request to attend another lab section during the same week (this may not be guaranteed as all lab sections has 25 students and are at full capacity)". If you miss lab during your assigned lab session you can not count on being able to make up your lab during another session time.

To be considered attending the lab session you must be present no later that 15 minutes after the start of the lab and you must be present the entire time

of the lab session unless approved and "checked off" by the lab Graduate Teaching Assistants that you are dismissed early.

Make-up Assignments for Authorized University Event or Co-curricular Activities

Students who represent the university at an authorized event or activity (for example, student athletes) and who are unable to meet a course deadline due to a conflict with that event must provide the instructor with documentation in advance to arrange a make-up. No penalty will be applied if the student gives advance notice and communicates with the instructor following UCF policy. In the case of an authorized university activity, it is your responsibility to show the instructor a signed copy of the Program Verification Form for which you will be absent, prior to the class in which the absence occurs or due date you need extended. For more information, see [UCF Policy 4-401](#).

Make-up Assignments for Religious Observances

A student who desires to observe a religious holy day of his or her religious faith must notify their instructor as soon as practicable prior to the observance. Students who are absent because of religious observances will be permitted a reasonable amount of time to complete any missed work. For more information, see [UCF Regulation 5-020](#).

Artificial Intelligence (AI) Use Policy

Use of Generative AI (GenAI) is freely permitted but must be disclosed. Students are allowed to use Generative Artificial Intelligence (GenAI) tools on assignments if the usage is properly disclosed and credited. For example, text generated from Copilot should include a citation such as: "Copilot. Accessed 2023-12-03. Prompt: 'Summarize the Geneva Convention in 50 words.' Generated using Copilot." Any attempt to represent GenAI output inappropriately as your own work will be treated as plagiarism. Ultimately, students are responsible for the accuracy and veracity of submitted work whether AI-generated or not.

Disability Access & Accommodations

The University of Central Florida is committed to providing equal access to all students with disabilities (ADHD, learning disabilities, Autism, chronic medical conditions, physical disabilities, etc.). To receive consideration for reasonable disability-related course accommodations, disabled students must contact Student Accessibility Services (SAS) and complete the steps required for SAS to review accommodation requests. More information can be found on the UCF [Student Accessibility Services](#) website under the Start Here tab or by contacting SAS directly (Ferrell Commons 185; sas@ucf.edu; Phone - 407-823-2371).

Approved accommodations are shared with course instructors via the SAS Course Accessibility Letter. Implementing certain accommodations may require discussion about specific considerations of the course design, course learning objectives, and the individual academic and course challenges experienced by the student. While students with disabilities or chronic health needs are also encouraged to discuss any course concerns with professors in addition to contacting SAS, professors are not required to facilitate disability-related adjustments to the course unless the professor has received a Course Accessibility Letter from SAS that outlines approved accommodations.

Academic Integrity

Students should familiarize themselves with UCF's Code of Conduct at Student Conduct and Integrity Office. According to Section 1, "Academic Misconduct," students are prohibited from engaging in:

1. Academic misconduct is defined as any submitted work or behavior that obstructs the instructor of record's ability to accurately assess the student's understanding or completion of course materials or degree requirements (e.g., assignment, quiz, and/or exam). Examples of academic misconduct include but are not limited to: plagiarism, unauthorized assistance to complete an academic exercise; unauthorized communication with others during an examination, course assessment, or project; falsifying or misrepresenting academic work; providing misleading information to create a personal advantage to complete course/degree requirements; or multiple submission(s) of academic work without permission of the instructor of record.

2. Any student who knowingly helps another violate academic behavior standards is also in violation of the standards.
3. Commercial Use of Academic Material. Selling of course material to another person and/or uploading course material to a third-party vendor without authorization or without the express permission of the University and the instructor of record. Course materials include but are not limited to class notes, the instructor of record's slide deck, tests, quizzes, labs, instruction sheets, homework, study guides, and handouts.
4. Soliciting assistance with academic coursework and/or degree requirements. The solicitation of assistance with an assignment, lab, quiz, test, paper, etc., without authorization of the instructor of record or designee is prohibited. This includes but is not limited to asking for answers to a quiz, trading answers, or offering to pay another to complete an assignment. It is considered Academic Misconduct to solicit assistance with academic coursework and/or degree requirements, even if the solicitation did not yield actual assistance (for example, if there was no response to the solicitation).

Responses to Academic Dishonesty, Plagiarism, or Cheating

Students should also familiarize themselves with the procedures for academic misconduct in UCF's student handbook, [The Golden Rule](#). UCF faculty members have a responsibility for students' education and the value of a UCF degree, and so seek to prevent unethical behavior and respond to academic misconduct when necessary. Penalties for violating rules, policies, and instructions within this course can range from a zero on the exercise to an "F" letter grade in the course. In addition, an Academic Misconduct report could be filed with the Office of Student Conduct and Academic Integrity, which could lead to disciplinary warning, disciplinary probation, or deferred suspension or separation from the University through suspension, dismissal, or expulsion with the addition of a "Z" designated on one's transcript.

Being found in violation of academic conduct standards could result in a student having to disclose such behavior on a graduate school application, being removed from a leadership position within a student organization, the recipient of scholarships, participation in University activities such as study abroad, internships, etc.

Let's avoid all of this by demonstrating values of honesty, trust, and integrity. No grade is worth compromising your integrity and moving your moral compass. Stay true to doing the right thing: take the zero, not a shortcut.

Title IX

Title IX prohibits sex discrimination, including sexual misconduct, sexual violence, sexual harassment, and retaliation. If you or someone you know has been harassed or assaulted, you can find resources available to support the victim, including confidential resources and information concerning reporting options at [Let's Be Clear](#) and [UCF Cares](#).

For more information on access and community engagement, Title IX, accessibility, or UCF's complaint processes contact:

- Title IX – ONAC – [Office of Nondiscrimination & Accommodations Compliance](#) & askanadvocate@ucf.edu
- Disability Accommodation – Student Accessibility Services – [Student Accessibility Services](#) & sas@ucf.edu
- [Access and Community Engagement](#) (including the Ginsberg Center for Inclusion and Community Engagement, Military and Veteran Student Success, and HSI Initiatives)
- UCF Compliance and Ethics Office – [Compliance, Ethics, and Risk Office](#) & complianceandethics@ucf.edu
- The [Ombuds Office](#) is a safe place to discuss concerns.

Reporting an Incident or Issue

If you believe you have experienced discrimination by any faculty or staff member, contact the Office of Nondiscrimination & Accommodations Compliance via the [ONAC website](#) or at 407-823-1336. You can also choose to report using the UCF Integrity Line either anonymously or as yourself at 1-855-877-6049 or by using the [online form](#). UCF cares about you and takes every report seriously. For more information see the [Reporting an Incident or Issue Webpage](#).

Class Schedule

Items	Topic
1	Syllabus, solder training, BOM for new msp430 project, and Autodesk license registration
2	TI WEBENCH, introduction to regulators and the testing of regulators.

Items	Topic
3	The student will prototype a 3.3 volt and a +5 volt regulator and an introduction to regulators.
4	The student will design and layout on a PCB using fusion360 these two regulators .
5	The student will design and layout on a PCB using fusion360 these two regulators and order their blank PCB's.
6	The student will design and layout on a PCB using fusion360 these two regulators and order their blank PCB's (suggest to be ordered with their junior design project by the end of week 10).
7	The student will research for the course project, and design Project PCB board schematic.
8	The student will research for the course project, and design Project PCB board schematic. The student will design PCB board layouts for their junior design msp430 project.
9	The student will design PCB board layouts for their junior design msp430 project. The students will order the PCB for their project.
10	The student will start the prototyping of their junior design project using a breadboard and then write the software for their junior design project.
11	The student will start the prototyping of their junior design project using a breadboard and then write the software for their junior design project.
12	Students will build and test their Regulator boards.
13	The students build their PCB and test their project.
14	Catchup as required.
15	Final Exam Week - Junior Design Project Demos and Discussion

Deployed Active-Duty Military Students

Students who are deployed active-duty military and/or National Guard personnel and require accommodation should contact their instructors as soon as possible after the semester begins and/or after they receive notification of deployment to make arrangements.

PROFESSIONALISM

Two (2) points will be deducted from the Overall Final Course Grade for each occurrence of the following:

§ IMPROPER/UNPROFESIONAL CONDUCT:

- Disturbances in class - verbally or electronically including the improper use of cellphones and electronic computer systems (using electronic computer systems other than for class lectures).
- Improper language used in the class lectures or emails or discussions in WEBCOURSES
- Criticizing another student in class for asking a technical question relative to the class material.
- ASKING TO TREAT ONE STUDENT DIFFERENTLY THAN ANOTHER STUDENT.
 - Asking the instructor to go over your solutions of assignments prior to submittal.
 - Asking to take an exam at a date different than the scheduled time given by the instructor or the university (except of university pre-approved activities).
 - Asking to turn in the project/assignment late after the due date.
 - Asking the instructor to grade the exam differently than the grading method used for all students in the class.

Any cheating on labs, pop quizzes, and exams will result in an immediate failure for the class. As reflected in the UCF creed, integrity and scholarship are core values that should guide our conduct and decisions as members of the UCF community. Plagiarism and cheating contradict these values, and so are very serious academic offenses. Penalties can include a failing grade in an assignment or in the course, or suspension or expulsion from the university. Students are expected to familiarize themselves with and follow the University's Rules of Conduct (see <https://scai.sswb.ucf.edu/academic-integrity/>)

Campus Safety

At UCF's Public Safety and Police, safety is the top priority. Emergencies on campus are rare, but if one should arise, it's important to be familiar with some basic safety and security concepts.

- In an emergency, always dial 911.
- Every UCF Classroom has an Emergency Procedure Guide posted on a wall near the door, which will show you how to respond to a variety of situations. This guide can also be found online [here](#).
- In the event of an active threat, remember **AVOID, DENY, DEFEND**. Choose the best course of action and act immediately. Watch the video [here](#) to learn more.
 - **AVOID**. Pay attention to your surroundings and have an exit plan. Get as much distance and as many barriers between you and the threat as quickly as possible.
 - **DENY**. When avoiding is difficult or impossible, deny the threat access to you and your space. Lockdown by creating barriers, turning the lights off and remaining quiet and out of sight. Make sure your phone is silenced, but do not turn it off.
 - **DEFEND**. When you are unable to put distance between yourself and the threat, be prepared to protect yourself. Commit to your actions, be aggressive and do not fight fairly. Do whatever it takes to survive.
- For emergencies on campus, UCF will utilize the [UCF Alert](#) system. All UCF students, faculty, and staff are automatically enrolled to receive these email and text alerts, however, it's a good idea to frequently ensure your [contact information is up to date](#).

Financial Aid Accountability

All instructors are required to document students' academic activity at the beginning of each course. In order to document that you began this course, please complete this activity by the end of the first week of classes or as soon as possible after adding the course. Failure to do so may result in a delay in the disbursement of your financial aid.